Allergic Conjunctivitis

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Introduction

- ☑ Ocular allergies occur in 15–25% of the population. The incidence seems to be increasing in recent years, as pollutant concentration in the ambient environment is increasing.
- ☑ Cornea, lens, anterior chamber, and vitreous are devoid of blood and lymph vessels. Inflammatory reactions occur mostly in the neighboring structure conjunctiva, which contains mast cells (90%).

Classification

- ☑ Seasonal and perennial atopic (rhino) conjunctivitis
- ☑ Vernal keratoconjunctivitis (VKC) or atopic keratoconjunctivitis (AKC)
- ☑ Giant papillary conjunctivitis (GPC)
- ☑ Allergic or hypersensitivity changes on the ocular surface.

Differential Diagnosis

Differential diagnosis includes any disease in which the eye becomes red with the symptoms of burning, itching, and tearing.

Infectious Causes

- ☑ *Viral conjunctivitis*: Starts with unilateral eye and later spreads to other eye within 3–4 days and often associated with mucopurulent discharge with preauricular lymphadenopathy.
- ☑ *Bacterial conjunctivitis*: Matting of the eyelids along with yellow to greenish purulent discharge is seen. Examination of the eye reveals a papillary reaction on the tarsal conjunctival surfaces and punctate keratopathy.
- ☑ *Chlamydial conjunctivitis*: Usually insidious onset and present at least 6 weeks before diagnosis made.

Noninfectious Causes

- ☑ Contact dermatitis
- ☑ Drug-induced allergic contact dermatitis
- ☑ Keratoconjunctivitis sicca
- ☑ Blepharitis
- ☑ Subconjunctival hemorrhage
- ☑ Contact dermatitis with conjunctivitis is mainly due to detergents, creams, lotions, cosmetics, or eye drops.

Treatment

- ☑ Main targets:
  - Treating the symptoms
  - Preventing degranulation
  - Treating the hyperemia

- ☑ Cold compresses
- ☑ Antihistamines—topical and systemic
- ☑ Tear substitutes—preservative free preferred
- ☑ Steroids—avoid long-term use to prevent glaucoma
- ☑ Nonsteroidal anti-inflammatory drugs (NSAIDs)—can reduce itching and hyperemia
- ☑ Mast cell stabilizers—best for long-term therapy—sodium cromoglycate (2–4%), nedocromil sodium, lodoxamide, olopatadine, azelastine, and epinastine
- ☑ Cyclosporine A and tacrolimus—topical application reduces the need for long-term steroids
- ☑ Antibiotics—in cases of suspected staphylococcal infection.

Seasonal/Perennial Allergic Conjunctivitis

Topical drops for allergic conjunctivitis come in many different forms. Here we outline a general overview/suggestions of the different options. Of note, chronic use of vasoconstrictive agents can lead to rebound vasodilation when discontinued.

- ☑ *First line*: Over-the-counter antihistamine/vasoconstrictor agent. Vasoconstrictors—inexpensive over-the-counter are first line. Many brands are available containing antazoline phosphate 0.05%, naphazoline HCl 0.05%, oxymetazoline HCl, tetrahydrozoline HCl 0.05%, or phenylephrine 0.12%.
- ☑ *Second line*: Second-generation topical histamine H1-receptor antagonist. H-1 receptor antagonists—more effective than the vasoconstrictors, but much more expensive. The three principal choices are pheniramine maleate 0.3%, emedastine, and levocabastine HCl 0.05%.
- ☑ *Third line*: Recurrent/persistent—mast cell stabilizers can be used. Mast cell stabilizers—cromolyn sodium 4%, nedocromil 2%, pemirolast 0.1%, and lodoxamide tromethamine 0.1%. These are Food and Drug Administration (FDA) approved only for a specifically vernal conjunctivitis. Use only if other classes of medications have failed.
- ☑ *Combined medications can also be used*: Combined H-1 receptor antagonist and mast cell stabilizers—olopatadine hydrochloride 0.1%, olopatadine hydrochloride 0.2%, optivar, ketotifen fumarate, and elestat. These agents are more effective than the mast cell stabilizers alone.
- ☑ *If symptoms are not adequately controlled*: Brief course of low-potency and low-frequency topical corticosteroids. NSAID such as ketorolac has been approved by the FDA for the use in seasonal allergic conjunctivitis but is generally used as adjunctive agent. Nonsteroidal anti-inflammatory agents—ketorolac tromethamine 0.5%.

Treatment

Seasonal/Perennial Allergic Conjunctivitis

- ☑ Oral antihistamines, such as levocetirizine, cetirizine, fexofenadine, desloratadine, and loratadine can be used, especially if the patient is experiencing additional systemic effects.
- ☑ Some additional measures can be used as adjunctive measures to improve symptom management, which are as follows:
  - Artificial tears to dilute allergens
  - Cool compresses/ice packs.

Avoiding Allergens

- ☑ Frequent clothes washing and bathing/showering before bedtime
- ☑ Refer/consult allergy or dermatology for those who are not adequately controlled with topical medications and oral antihistamines.

Vernal/Atopic Keratoconjunctivitis

- ☑ Although the disease process is more advanced in both VKC and AKC, the general treatment strategies remain similar to management of seasonal or perennial allergic conjunctivitis. If, however, an acute exacerbation occurs, topical corticosteroids are often added to reduce inflammation.
- ☑ Another adjunctive therapy is addition of topical cyclosporine 2%. If the VKC or AKC is not responsive to topical therapy, supratarsal injection of corticosteroid can be considered.

Giant Papillary Conjunctivitis

- ☑ The most important management in GPC is modifying the causative entity.
- ☑ This generally includes removal or replacement of sutures, rotating the knots, or using a therapeutic contact lens.

Further Reading

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